

Airline Booking Funnel Analysis



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Where do we lose customers, and is it worse for any channel or trip type?



Problem

Airlines lose a large share of revenue between a customer starting a search and completing payment. This analysis identifies where in the booking flow that drop-off is worst, and whether it varies by sales channel or trip type — using a dataset of ~50,000 real airline booking attempts.

Approach

Analyzed the full dataset in SQL to calculate conversion rates overall and by segment. Built an interactive Tableau dashboard to visualize conversion by sales channel, trip type, and route. Reshaped a randomly sampled subset of the data into event-level records and built a segmented conversion funnel in Mixpanel to validate the findings in a dedicated product analytics tool.

What I Found

Mobile converts ~30% lower than Internet (10.84% vs. 15.48%), confirmed independently across SQL, Tableau, and Mixpanel — and the gap holds even after controlling for trip type, so it isn't explained by what people are booking, likely by how they're booking on mobile.

OneWay and CircleTrip bookings are under 1% of total volume with lower conversion, consistent with users price-checking one-way fares to benchmark round-trip pricing rather than intending to fly one-way — a low-priority segment given its size.

One route (DELSYD) showed 0% conversion across 100+ booking attempts — statistically unlikely to be organic low demand, and flagged as worth investigating as a possible booking-flow or data issue.

What I'd do Next

If Mobile converted at Internet's rate, this dataset alone would represent roughly 261 additional completed bookings (5,618 mobile attempts × the 4.6-point conversion gap) — and if mobile traffic share continues growing, that gap compounds over time. I'd propose an A/B test on the mobile checkout flow: split incoming mobile traffic into a control (current flow) and a variant (simplified checkout — fewer steps, larger tap targets, saved payment details), and measure conversion lift over a 2–4 week window with a pre-calculated minimum sample size. If the variant closes even half the gap, that's roughly 130 additional bookings recovered from existing mobile traffic with no added acquisition spend.

Methodology Note: An initial 20,000-row sample (taken as the top rows of the dataset) produced a skewed conversion rate (~11.2% blended vs. the true ~15%), traced to non-random ordering in the source data. Corrected by shuffling with a random sort key before resampling; results now align closely across SQL, Tableau, and Mixpanel.

Tools: SQL (BigQuery) • Tableau Public • Mixpanel | Dataset: British Airways customer booking data (Kaggle, ~50,000 rows)